Supporting Information: Reservoir Computing with Quorum-Sensing Genetic Oscillators for Biosignal Classification

James Newsome, Tim Rudge, and Gizem Buldum*

Interdisciplinary Computing and Complex Biosystems Research Group, School of Computing, Newcastle University, Newcastle upon Tyne, NE1 7RU, United Kingdom

E-mail: gizem.buldum@newcastle.ac.uk

Contents

1 Gene Expression in Quorum-Coupled Bio-Pixels

1.1 Parameter Values

Table S1: Model parameters adopted from Danino et al. (S1).

Parameter	Description	Value
C_A	AiiA production rate	1.0
C_I	LuxI production rate	4.0
δ	Basal promoter activity	1×10^{-3}
α	Promoter activation strength	2500.0
k	Saturation constant (LuxI)	1.0
k_1	Promoter half-saturation	0.1
b	AHL synthesis scaling	0.06
γ_A	AiiA degradation rate	15.0
γ_I	LuxI degradation rate	24.0
γ_H	AHL degradation rate	0.01
f	Protein interaction coefficient	0.3
g	AHL–AiiA interaction coefficient	0.01
ρ	Dilution parameter	0.7
ρ_0	Critical dilution threshold	0.88
D	Diffusion rate constant	2.5
μ	Extracellular decay rate	0.6
τ	Delay (timesteps)	10

1.2 Convergence Analysis

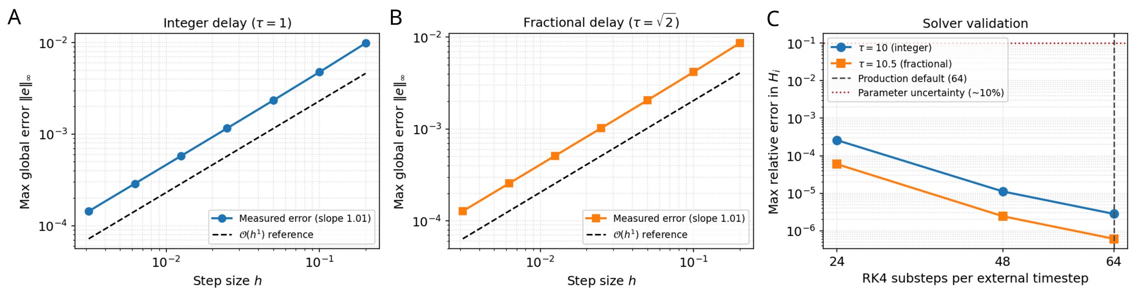

Figure S1: Convergence and solver validation of the frozen-delay numerical integration (RK4) scheme. (a) Convergence on a scalar DDE with integer delay ($\tau = 1$) showing first-order global accuracy ($\mathcal{O}(h^1)$). (b) Convergence with fractional delay ($\tau = \sqrt{2}$), confirming the same first-order rate when the interpolation code path is exercised. (c) Production solver validation on the biological oscillator model: accumulated relative error in H_i over 50 timesteps for 24, 48, and 64 RK4 substeps relative to a 96-substep reference. At the production value of 64 substeps, the relative error remains well below 10^{-4}.

2 Dataset Characteristics and Preprocessing

2.1 Dataset Overview

Table S2 shows the raw class distribution, and Table S3 reports approximate per-fold instance counts for all three task configurations under stratified five-fold CV. All stochastic procedures used a fixed random seed (6337) for reproducibility.

Table S2: Raw class distribution in the MIT-BIH dataset with labeling from Kachuee (S2, S3).

Class	Count	Percentage
Normal (N)	90 589	82.8%
Supraventricular (S)	2 779	2.5%
Ventricular (V)	7 236	6.6%
Fusion (F)	803	0.7%
Unknown (Q)	8 039	7.3%
Total	109 446	100%

Table S3: Approximate per-fold instance counts for all three tasks under stratified five-fold cross-validation. Values are approximate as exact counts vary by ±1 across folds due to integer partitioning.

	Balanced 5-class		Unbalanced 5-class		Binary	
Class	Train	Test	Train	Test	Train	Test
N	~642	~161	3 340	835	~15 086	~3 771
S	~642	~161	~2 223	~556	—	—
V	~642	~161	3 340	835	—	—
F	~642	~161	~642	~161	—	—
Q	~642	~161	3 340	835	—	—
Arr.	—	—	—	—	~15 086	~3 771
Total	3 212	803	~12 885	~3 222	~30 172	~7 542
CV pool	4 015		16 107		37 714	

2.2 Normalization

Per-sequence z-score normalization was applied in all experiments:

$$\tilde{x}_t = \frac{x_t - \mu_x}{\sigma_x}, \tag{1}$$

where μ_x and σ_x are the mean and standard deviation of the individual sequence. The selection of per-sequence z-score over five alternative strategies was validated by the Phase 1 normalization study (Table S7).

2.3 Noise Augmentation

For each selected training sample, the augmentation procedure generated a noisy copy as:

$$x_t^{\mathrm{aug}} = x_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \eta^2 \sigma_x^2), \tag{2}$$

where η is the noise rate (fraction of per-sequence standard deviation used as noise amplitude) and σ_x is the standard deviation of the original sequence.

Additive Gaussian noise augmentation was evaluated as part of the Phase 1 sensitivity analysis (P1-noise-data, Table S6). At the Phase 1 operating point ($N = 250$), the best augmentation parameters were `noise_rate` $= 0.300$ and `noise_ratio` $= 0.286$ ($F_1 = 0.803$), both near the upper boundary of their search range.

To validate this finding at the final operating point ($N = 1000$, Phase 2 best-trial configuration), we conducted a dedicated $7 \times 7 = 49$-point grid search over noise rate (0–0.3) and augmentation ratio (0–0.3) using five-fold cross-validation. The grid showed negligible sensitivity to augmentation: the best setting (rate = 0.15, ratio = 0.30) achieved $F_1 = 0.884$, indistinguishable from the non-augmented baseline ($F_1 = 0.884$) at three decimal places. The full F_1 landscape spanned a range of just 0.003 across all 49 grid points. This result is consistent with the flat F_1 landscapes observed in the Phase 1 noise studies (Figure S5). Based on this evidence, noise augmentation was excluded from all headline evaluations.

3 Classification Workflow and Algorithmic Implementation

3.1 Cross-Validation Procedure

The stratified five-fold classification procedure described in the main text is summarized in pseudocode below:

```
CROSS-VALIDATE(X, Y, reservoir, readout, k):
    folds = stratified_k_fold_split(X, Y, k)
    fold_metrics = []

    FOR EACH (X_train, Y_train, X_val, Y_val) IN folds:
        X_train, X_val = normalise(X_train, X_val)

        train_states = reservoir.run(X_train)
        readout.fit(train_states, Y_train)
```

```
        val_states = reservoir.run(X_val)
        Y_pred = readout.predict(val_states)

        fold_metrics.append(evaluate(Y_val, Y_pred))

    RETURN mean(fold_metrics)
```

3.2 Reservoir State Extraction

Each ECG instance was processed independently:

```
RESERVOIR.RUN(X):
    states = []
    FOR EACH instance IN X:
        reset_reservoir_state()
        state = integrate_dde(reservoir_weights, instance, dt)
        states.append(final_state)
    RETURN states
```

4 Optimization Phase 1: Sensitivity Analysis

4.1 Optimization Framework and Computational Setup

Hyperparameter optimization used Bayesian exploration with Optuna (S4) distributed across five servers (Table S4), each running Linux RHEL 9 with 12–64 asynchronous workers sharing a single SQLite database per study. Macro F1 served as the primary objective for all studies and each Phase 1 trial used a single stratified fold as a proxy metric.

All experiments used Python 3.9.21 (key packages: reservoirpy 0.3.11, scikit-learn 1.5.0, NumPy 1.26.4, Optuna 3.6.1, Numba 0.60.0). Pinned versions are tracked in a `pyproject.toml` lockfile.

Table S4: Hardware configuration of the five servers used for hyperparameter optimization. Studies were allocated half of their host's logical cores for workers, and ran in parallel across hosts.

Host	Processor	Physical Cores	Memory (GB)
Host 1	Intel Xeon Gold 6538Y+ (2×)	64	1511
Host 2	Intel Xeon Silver 4516Y+ (2×)	48	754
Host 3	AMD EPYC 7302 (2×)	32	755
Host 4	Intel Xeon Silver 4410Y (2×)	24	251
Host 5	Intel Xeon Silver 4410Y	12	125

4.2 Studies Overview

Studies use a phase prefix (`P1`–`P4`) plus a descriptive suffix. Abbreviated forms (e.g. `P1-topo-dist`) appear in tables where space is limited. Per-trial resource usage at $N = 250$ was approximately 400 MB RSS, with wall-clock times of 430 s–1250 s under concurrent worker load. A shared baseline was held fixed except for the parameter group under investigation and updated after Phase 1 (Table S6).

Table S5: Complete optimization study overview across the four BioReservoir phases, as well as the ESN and no-reservoir baselines, using Macro F1 as the primary objective. Phase 1 studies isolate individual parameter groups (single-fold proxy), and Phase 2 jointly optimizes all important parameters (single-fold). Phase 3 evaluates the best configuration under alternative dataset settings with five-fold CV, and Phase 4 optimizes the readout classifier on frozen reservoir states, again with five-fold CV. The ESN studies apply the same training/readout workflow to a standard leaky ESN baseline. The Baseline studies train the same classifier families (plus MLP and perceptron) directly on raw z-scored ECG features without any reservoir transformation, using five-fold CV on the three headline dataset configurations.

Phase	Study	Host	Parameter Group	Sampler	Trials	Folds	Best F1
1	P1-scaler	5	Data normalization	Grid	6	1	0.760
	P1-scaling	2	Signal scaling	TPE	288	1	0.896
	P1-integration	3	DDE integration	TPE	128	1	0.816
	P1-topo-dist	1	Distance topology	TPE	256	1	0.836
	P1-topo-rand	1	Random topology[a]	TPE	256	1	0.786
	P1-noise-data	5	Data augmentation	TPE	96	1	0.803
	P1-noise-res	4	Reservoir noise	TPE	144	1	0.795
	P1-input	4	Input layer	TPE	192	1	0.865
	P1-readout	2	Output aggregation	TPE	191	1	0.783
	P1-units	2	Reservoir size[b]	Grid	10	1	0.895
	P1-coupling	3	Cell coupling	TPE	128	1	0.896
2	P2-distance	1	Joint (distance)	TPE	768	1	0.905
	P2-random	2	Joint (random)	TPE	800	1	0.907
3	P3-augmentation	1	Augmentation grid	Grid	49	5	0.884
	P3-imbalance	1	Class imbalance grid	Grid	5	5	0.904
	P3-binary	1	Binary classification	—	1	5	0.931
4	P4-readout-bal.	4	Balanced readout	TPE	950	5	0.895
	P4-readout-unbal.	2	Unbalanced readout	TPE	939	5	0.916
	P4-readout-bin.	3	Binary readout	TPE	974	5	0.933
ESN	ESN-balanced	2	Leaky ESN training	TPE	240	1	0.879
	ESN-readout	4	Frozen-state readout	TPE	1008	5	0.896
Baseline[c]	B1-balanced	2	Balanced 5-class	TPE	480	5	0.886
	B2-unbalanced	2	Unbalanced 5-class	TPE	480	5	0.915
	B3-binary	2	Binary	TPE	480	5	0.935

Table S6: Phase 1 best-trial parameter values adopted as the updated baseline for Phase 2 joint refinement. Baseline values are from preliminary iterative experiments. The best values were identified by the corresponding sensitivity study.

Parameter	Study	Baseline	Best Value	Search Range
`scaler_type`	P1-scaler	`seq_zscore`	`seq_zscore`	6 strategies
`input_scaling`	P1-scaling	8.01×10^{-2}	1.03×10^{-3}	$[10^{-4}, 0.5]$
`rc_scaling`	P1-scaling	1.95×10^{-2}	5.71×10^{-4}	$[10^{-4}, 0.2]$
`dde_scaling`	P1-scaling	3.70×10^{-3}	7.46×10^{-3}	$[5 \times 10^{-5}, 0.02]$
`rk4_substeps`	P1-integration	44	60	[12, 64]
`warmup`	P1-integration	40	50	[10, 80]
`sparsity`	P1-topo-dist	0.50	0.134	[0.05, 1.0]
`fade_alpha`	P1-topo-dist	1.81×10^{-5}	1.01×10^{-6}	$[10^{-7}, 10^{-2}]$
`p_excite`	P1-topo-dist	0.50	0.66	[0.3, 0.7]
`signed`	P1-topo-dist	False	True	{True, False}
`cell_coupling`	P1-coupling	16.28	8.86	[0.01, 100]
`input_connectivity`	P1-input	0.19	0.368	[0.1, 1.0]
`output_variables`	P1-input	`IHe`	`He`	6 subsets
`readout_aggregation`	P1-readout	`final`	`last_N`	3 strategies
`readout_window`	P1-readout	1	36	[1, 200]
`noise_in`	P1-noise-res	7.87×10^{-2}	1.44×10^{-2}	[0, 0.3]
`noise_rc`	P1-noise-res	0.121	0.240	[0, 0.3]
`noise_rate`	P1-noise-data	8.25×10^{-2}	0.300	[0, 0.3]
`noise_ratio`	P1-noise-data	0.235	0.286	[0, 0.3]

4.3 Dataset Normalization (P1-scaler)

Table S7: Classification Macro F1 performance for a grid search of six normalization strategies. F1-scores confirm the clear advantage of per-sequence z-score normalization, slightly exceeding global z-score normalization, but greatly exceeding all varieties of min-max. The relatively low F1 values here reflect the pre-optimization reservoir configuration performance.

Normalization Strategy	Macro F1
`sequence_zscore` (per-beat z-score)	0.760
`standard` (global z-score)	0.756
`minmax` (global min-max)	0.692
`none` (no normalization)	0.690
`sequence_minmax` (per-beat min-max)	0.681
`sequence_minmax_pos` (per-beat min-max, non-negative)	0.665

4.4 Signal Amplitude Scaling (P1-scaling and P1-coupling)

Table S8: Search ranges and best trial values for signal scaling and coupling parameters. All continuous parameters were sampled on a log scale using TPE. Best F1 is reported per single-fold study.

Parameter	Study	Search Range	Scale	Best Value	Best F1
`input_scaling`	Phase 1 scaling	$[10^{-4},\ 0.5]$	log	1.03×10^{-3}	
`rc_scaling`	Phase 1 scaling	$[10^{-4},\ 0.2]$	log	5.71×10^{-4}	0.896
`dde_scaling`	Phase 1 scaling	$[5 \times 10^{-5},\ 2 \times 10^{-2}]$	log	7.46×10^{-3}	
`cell_coupling`	Phase 1 coupling	$[10^{-2},\ 10^{2}]$	log	8.86	
`input_scaling`	Phase 1 coupling	$[10^{-4},\ 10^{-1}]$	log	5.65×10^{-2}	0.896
`rc_scaling`	Phase 1 coupling	$[10^{-4},\ 10^{-1}]$	log	2.78×10^{-2}	
`dde_scaling`	Phase 1 coupling	$[10^{-4},\ 10^{-2}]$	log	4.32×10^{-4}	

Both studies confirmed a low-amplitude operating regime, with coupling–scaling interactions (top trials at `cell_coupling` $\in [8, 30]$) motivating joint refinement in Phase 2. Performance was primarily governed by the ratio α_u/α_r rather than absolute magnitudes. The perturbation coefficient λ was effective in the range $[2 \times 10^{-3}, 8 \times 10^{-3}]$; values above 10^{-2} disrupted oscillatory stability.

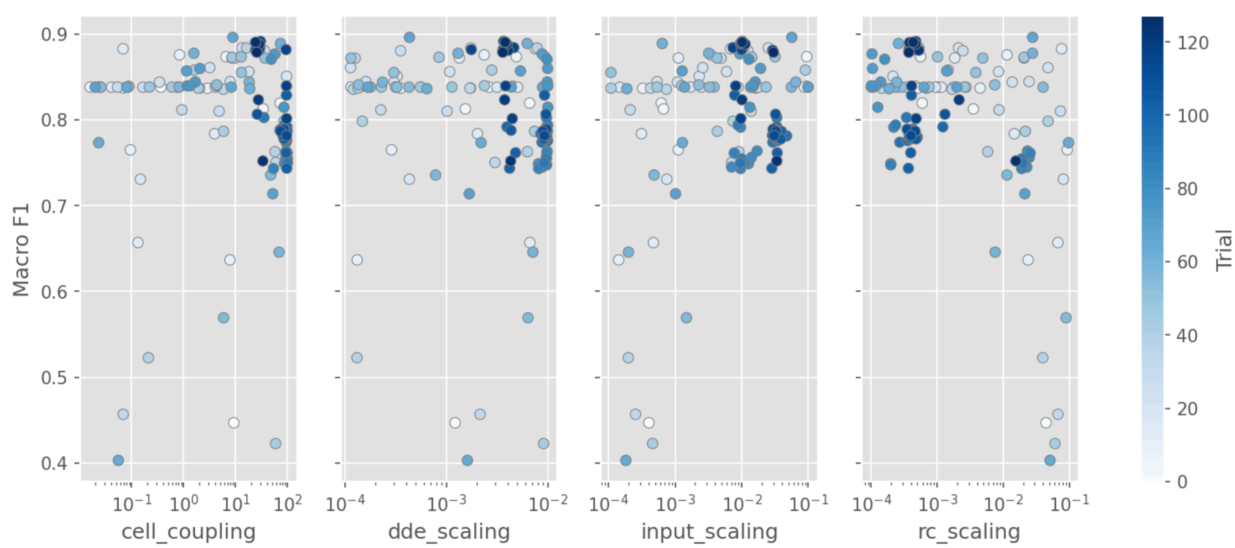

Figure S2: Slice plot for the Phase 1 coupling study. Joint variation of coupling strength with the three scaling coefficients reveals a broad high-performing basin at `cell_coupling` $\in [8, 30]$, confirming that interactions between coupling and scaling motivate joint refinement in Phase 2.

4.5 DDE Integration Parameters (P1-integration)

The best trial achieved $F_1 = 0.816$ at `rk4_substeps` $= 60$ and `warmup` $= 50$, both at the upper boundary of their search ranges. The `rk4_substeps` ceiling was accordingly extended to 88 for Phase 2.

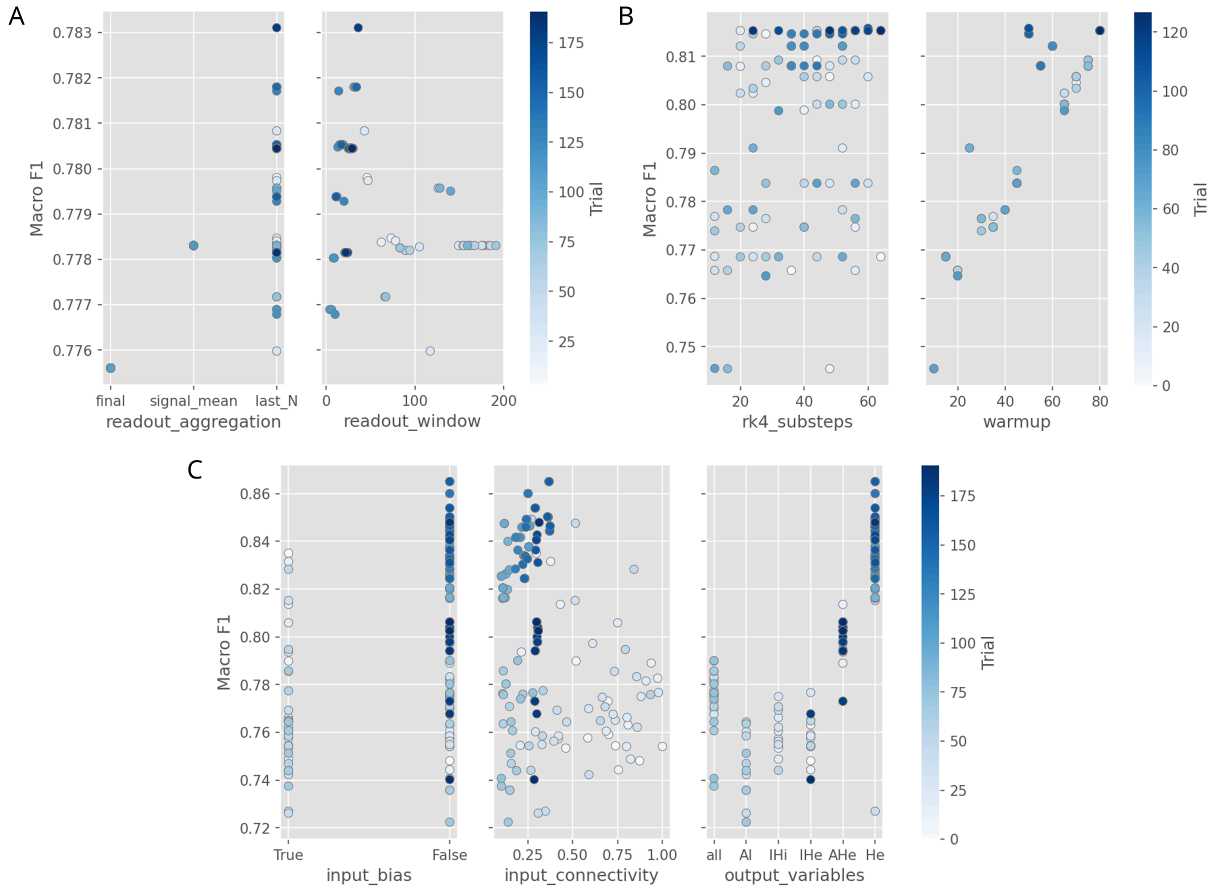

Figure S3: Phase 1 sensitivity slice plots for readout aggregation, DDE integration, and input-layer studies. (a) P1-readout shows that temporal aggregation over the last 36 timesteps substantially outperforms single-timestep readout. (b) P1-integration reaches the upper boundary of the search range, prompting an extended ceiling in Phase 2. (c) P1-input shows that H_e alone outperforms multi-variable state representations at $N = 250$.

4.6 Reservoir Topology (P1-topology-distance and P1-topology-random)

The distance topology achieved $F_1 = 0.836$ vs. 0.786 for random, though we must note that all fixed parameters were based on preliminary results with a distance-topology setup. An unbiased comparison is provided by Phase 2.

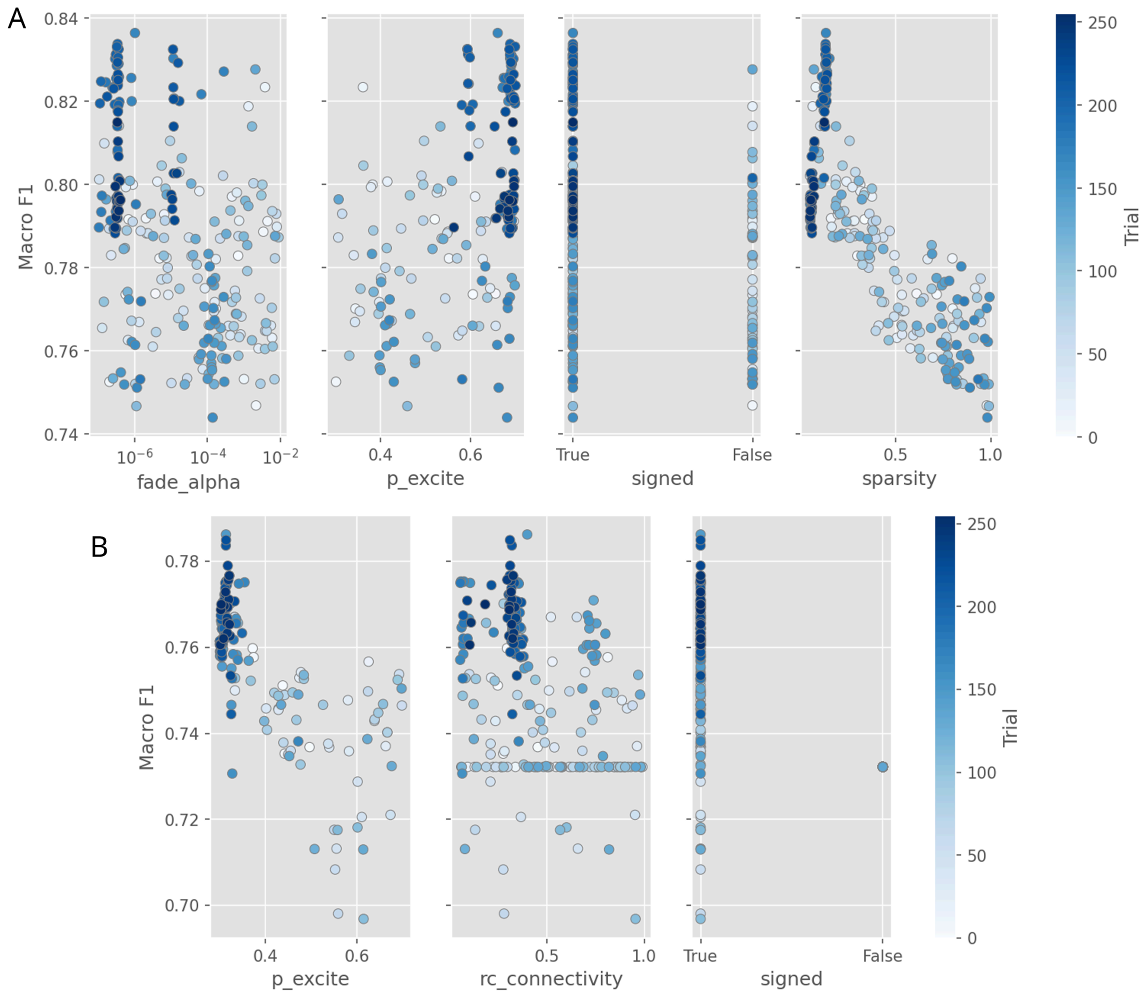

Figure S4: Slice plots for reservoir topology studies. (a) The distance-based topology favors sparse connectivity (`sparsity` $= 0.134$) with slow spatial decay. (b) The random topology achieves lower peak F1 under Phase 1 conditions, but this gap narrows under independent joint optimization in Phase 2.

4.7 Input Layer and Output Variables (P1-input)

Best values were `input_connectivity` = 0.368, `input_bias` = False, and `output_variables` = `He` ($F_1 = 0.865$). At $N = 250$, H_e alone outperformed higher-dimensional state vectors. Phase 2 joint optimization later selected $\{I, H_e\}$ at $N = 1000$, reflecting a parameter interaction where the larger reservoir size benefits from the additional feature dimensions.

4.8 Readout Aggregation Strategy (P1-readout)

`last_N` aggregation with $K = 36$ ($F_1 = 0.783$) outperformed both `signal_mean` and `final` (single-timestep) readout. The baseline was updated accordingly for Phase 2, earlier Phase 1 studies used the suboptimal `final` aggregation so their absolute F1 values in Table S5 underestimate Phase 2-achievable performance.

4.9 Data Augmentation and Reservoir Noise (P1-noise-data and P1-noise-reservoir)

Best augmentation parameters were `noise_rate` = 0.300 and `noise_ratio` = 0.286 ($F_1 = 0.803$), both near the upper boundary. Reservoir noise best values were `noise_in` = 0.014 and `noise_rc` = 0.240 ($F_1 = 0.795$). Both noise studies demonstrated mild patterns, with a maximum F1 range of ≈ 0.04 in the P1-noise-data study, and ≈ 0.07 in the P1-noise-reservoir study, confirming that noise parameters are mostly weak levers in this architecture, apart from σ_i which clearly degrades performance as it increases. A Phase 3 augmentation grid search at the final $N = 1000$ operating point subsequently confirmed that data augmentation was negligible (Section 2.3) and was therefore excluded from all headline evaluations.

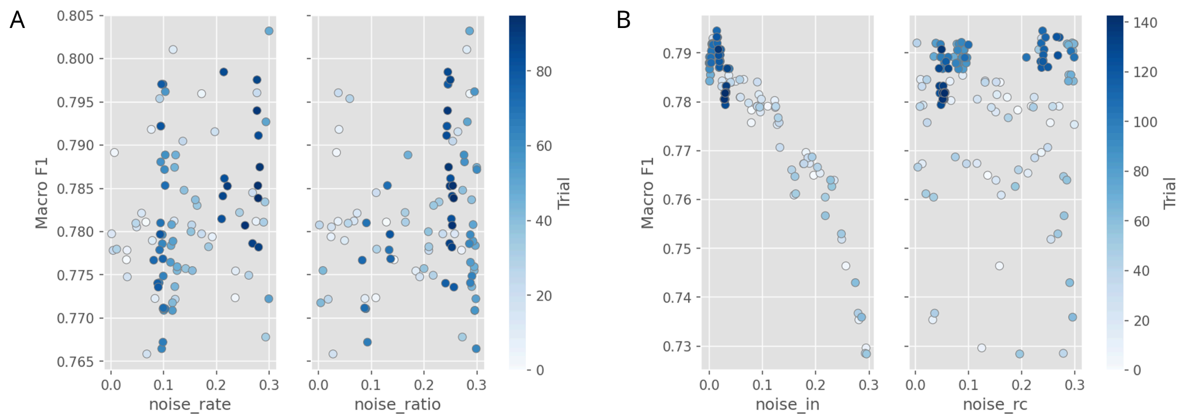

Figure S5: Slice plots for two noise studies which work separately at the dataset and reservoir levels. (a) Data augmentation rate and amplitude both reach the upper boundary, with mild patterns depicting performance improvement with more noise. (b) Reservoir noise studies show modest recurrent noise ($\sigma_r \approx 0.24$) as mildly beneficial. However, high reservoir input noise clearly degrades performance, with a peak value of $\sigma_u = 0.014$.

4.10 Reservoir Size Scaling (P1-units)

Reservoir size N was varied from 50 to 1000 nodes while holding all other parameters at the Phase 2 distance-topology best-trial values. Classification accuracy plateaued beyond $N \approx 300$ (Figure S6).

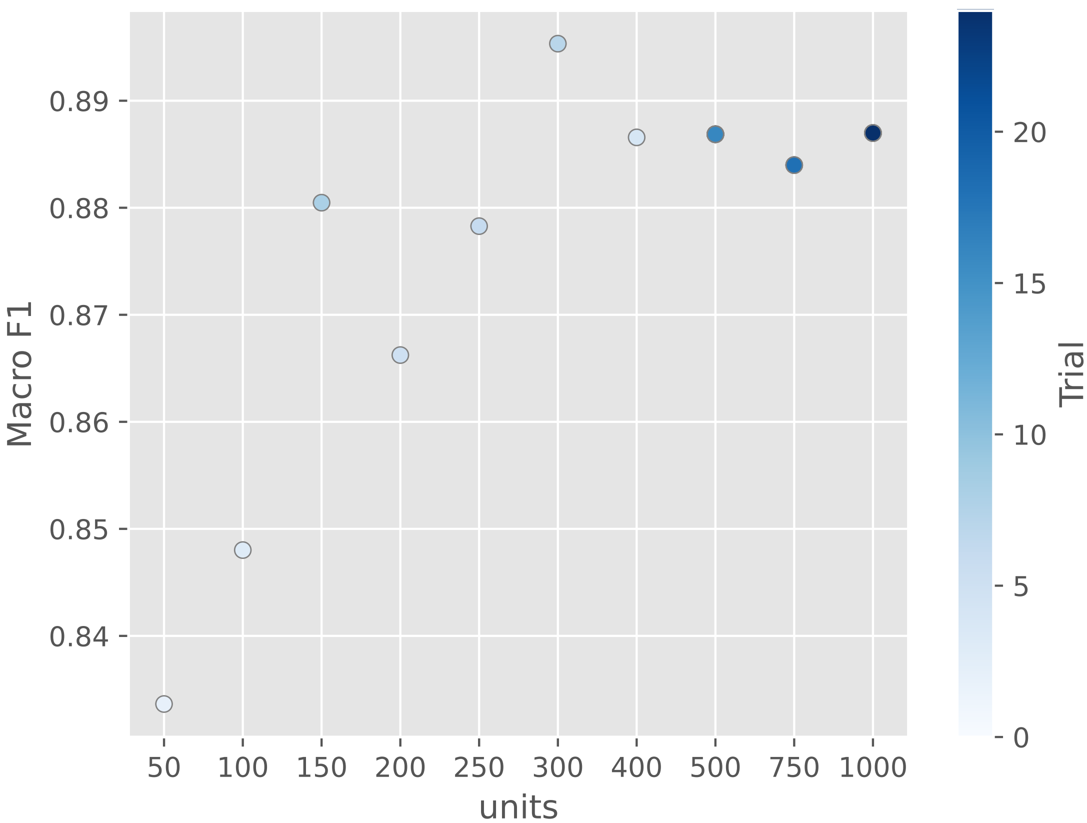

Figure S6: Phase 1 sensitivity of classification performance to reservoir size (P1-units study). Macro F1 peaks around $N = 300$ and then remains on a broad plateau across larger reservoirs, indicating diminishing returns beyond moderate reservoir sizes. Notably, this study used the best parameters from the other Phase 1 studies to ensure its results are more representative.

5 Optimization Phase 2: Joint Refinement

Phase 2 jointly optimized all influential parameters via TPE on a single stratified fold (768 completed trials for P2-distance, 800 for P2-random). This eliminates the Phase 1 confounding from a distance-topology baseline prior. Both phases shared the same fold partition, so Phase 3 five-fold CV provides the only held-out generalization estimate.

5.1 Distance-Based Topology (P2-distance)

Table S9: Best-trial parameter values from Phase 2 joint refinement (Best score: $F_1 = 0.905$).

Parameter	Value	Phase 1 Baseline
`units`	1000	250
`input_scaling`	2.77×10^{-2}	1.03×10^{-3}
`rc_scaling`	7.04×10^{-3}	5.71×10^{-4}
`dde_scaling`	6.44×10^{-4}	7.46×10^{-3}
`cell_coupling`	12.75	8.86
`rk4_substeps`	64	60
`warmup`	50	50
`sparsity`	0.576	0.134
`fade_alpha`	2.8×10^{-5}	1.01×10^{-6}
`p_excite`	0.835	0.66
`input_connectivity`	0.528	0.368
`output_variables`	`IHe`	`He`
`readout_aggregation`	`signal_mean`	`last_N`$(K = 36)$

Notable shifts from Phase 1: reservoir size increased to $N = 1000$, readout aggregation switched to `signal_mean`, and output variables changed to $\{I, H_e\}$.

5.2 Random Gaussian Topology (P2-random)

The best trial achieved $F_1 = 0.907$ (Table S10). The random topology converged to a qualitatively different regime: smaller reservoir ($N = 250$) with near-full connectivity (0.900). Both topologies independently selected `signal_mean` aggregation and `IHe` output variables.

Table S10: Best-trial parameter values from Phase 2 joint refinement (Best score: $F_1 = 0.907$).

Parameter	Value	Phase 1 Baseline
`units`	250	250
`input_scaling`	1.58×10^{-3}	1.03×10^{-3}
`rc_scaling`	5.80×10^{-5}	5.71×10^{-4}
`dde_scaling`	1.52×10^{-2}	7.46×10^{-3}
`cell_coupling`	6.97	8.86
`rk4_substeps`	72	60
`warmup`	30	50
`rc_connectivity`	0.900	0.397
`p_excite`	0.457	0.314
`input_connectivity`	0.965	0.368
`output_variables`	`IHe`	`He`
`readout_aggregation`	`signal_mean`	`last_N`$(K = 36)$

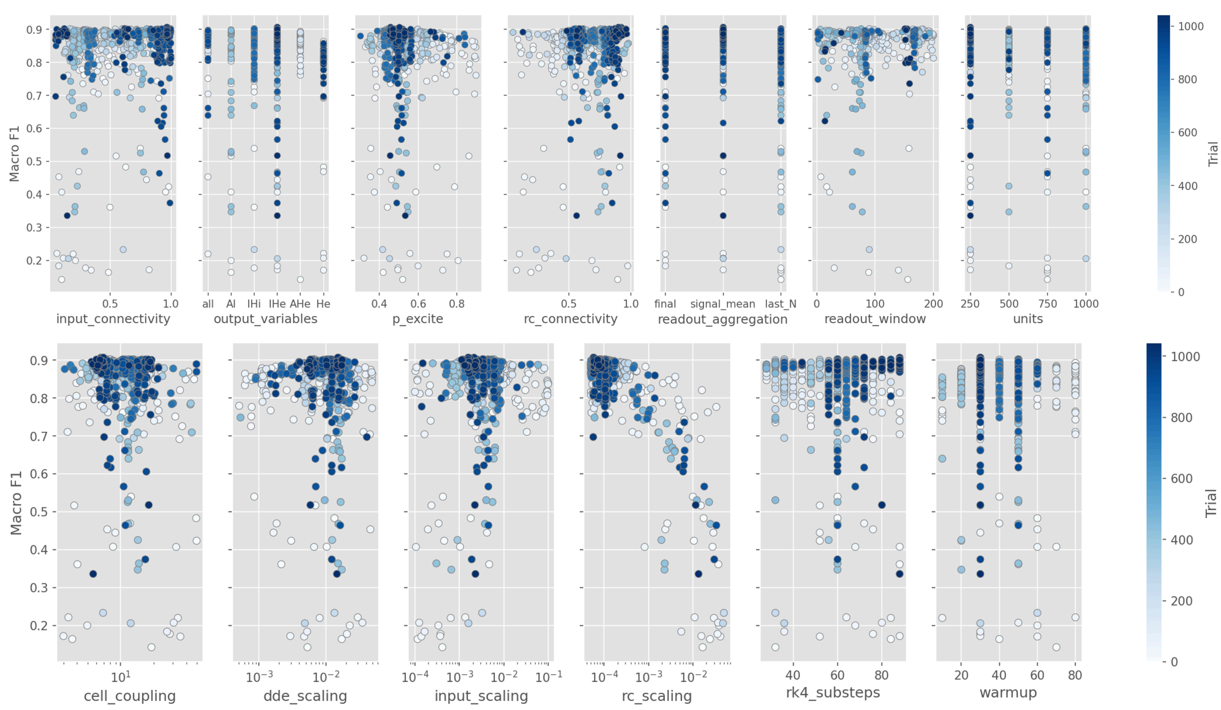

Figure S7: Phase 2 joint optimization for the random Gaussian topology (P2-random, 800 trials). The random topology converges to a qualitatively different parameter regime than the distance-based topology, with near-full recurrent and input connectivity, higher DDE scaling, and a smaller preferred reservoir size. The upper row summarizes topology, input, and readout parameters, the lower row summarizes scaling and integration parameters.

5.3 Topology Comparison Under Robust Evaluation

Under five-fold CV the distance topology generalized better despite marginally lower single-fold performance (Table S11), suggesting that P2-random was more sensitive to the optimization fold. Both studies used a baseline KNN ($k = 4$) readout; the Phase 4 readout optimization (Section 7) was performed only on the distance topology.

Table S11: Performance comparison between distance-based (P2-distance) and random Gaussian (P2-random) topologies. Single-fold values are from the best optimization trial, five-fold values were obtained by re-running the best trial with 5 fold cross-validation.

Metric	Distance (P2-distance)	Random (P2-random)
Single-fold F_1 (optimization)	0.905	0.907
Five-fold F_1 (mean $\pm$ std)	0.890 ± 0.007	0.881 ± 0.015
Five-fold accuracy	0.890	0.880
Five-fold MCC	0.863	0.851
Five-fold κ	0.862	0.850

6 Optimization Phase 3: Dataset Configurations

Phase 3 evaluated the best Phase 2 distance-topology configuration (P2-distance) under three additional experimental settings. The first is a dedicated augmentation grid search to re-assess its effect on performance under the optimized reservoir configuration (reported in Section 2.3). The remaining two studies use different dataset configurations (imbalance, binary).

6.1 Class Imbalance Study (P3-imbalance)

Maximum samples per class were varied from $1\times$ (803) to $5\times$ (4015) - each class contributes up to the cap or its available count. Table S12 shows the five-fold cross-validation results, where headline macro F1 at $5\times$ is 0.911 ± 0.004 (main text, Table 3).

Table S12: Macro F1-score as a function of the per-class training cap (five-fold stratified cross-validation, $N = 1000$, KNN $k = 4$ readout, no noise augmentation). Each class contributes up to the cap or its available count, so some classes do not reach the cap, producing class imbalance.

Multiplier	Max per Class	CV Pool Size	Macro F1
1×	803	4 655	0.887
2×	1 606	7 867	0.894
3×	2 409	11 079	0.901
4×	3 212	13 698	0.901
5×	4 015	16 107	0.904

6.2 Binary Classification (P3-binary)

Using KNN ($k = 4$) on a 4,015-instance readout-selection subset, the model achieved $F_1 = 0.931 \pm 0.005$. No optimization study was required as we just evaluated the existing best distance-topology reservoir parameters on a binary dataset configuration.

7 Optimization Phase 4: Readout Optimization

Phase 4 optimized the readout classifier on frozen reservoir states from Phases 2–3. Seven classifier families were evaluated via Optuna TPE with five-fold CV and macro F1 as the objective.

7.1 Balanced Five-Class Readout (P4-readout-balanced)

Table S13: Best macro F1-score per classifier for the balanced five-class readout study (950 trials, fivefold CV on frozen reservoir states).

Classifier	Best F1	Key Hyperparameters
Random Forest	**0.895**	$n_{\text{est}} = 326$, criterion = log_loss, depth = 23
Gradient Boosting	0.887	$n_{\text{est}} = 478$, lr = 0.83, loss = exponential
KNN	0.883	$k = 4$, weights = distance, $p = 1.03$
Ridge Classifier	0.862	$\alpha = 70.0$, solver = svd
Logistic Regression	0.862	$C = 0.39$, solver = liblinear
SVM	0.850	$C = 8 \times 10^{-5}$, kernel = linear
Decision Tree	0.823	criterion = log_loss, leaf = 5

7.2 Unbalanced Five-Class Readout (P4-readout-unbalanced)

Table S14: Best macro F1-score per classifier for the unbalanced five-class readout study (P4-readout-unbalanced, 939 trials, five-fold CV on frozen reservoir states).

Classifier	Best F1	Key Hyperparameters
Random Forest	**0.916**	$n_{\text{est}} = 321$, criterion = entropy, depth = 23
Gradient Boosting	0.897	$n_{\text{est}} = 107$, lr = 0.24
KNN	0.891	$k = 1$, weights = uniform, $p = 1.61$
Ridge Classifier	0.880	$\alpha = 2.8$
Logistic Regression	0.878	$C = 8.7$, solver = saga
SVM	0.849	$C = 9 \times 10^{-4}$, kernel = poly, degree = 2
Decision Tree	0.849	criterion = log_loss, leaf = 3

7.3 Binary Readout (P4-readout-binary)

Table S15: Best macro F1-score per classifier for the binary readout study (P4-readout-binary, 974 trials, evaluated on a 4,015-instance subset). The final headline binary result ($F_1 = 0.961$) was obtained by applying the selected readout to the full balanced dataset.

Classifier	Best F1	Key Hyperparameters
Random Forest	**0.933**	$n_{\text{est}} = 279$, criterion = log_loss, depth = 27
Gradient Boosting	0.931	$n_{\text{est}} = 415$, lr = 0.41
KNN	0.929	$k = 3$, weights = distance, $p = 1.73$
Ridge Classifier	0.921	$\alpha = 2.8$, solver = sparse_cg
Logistic Regression	0.917	$C = 1448$, solver = newton-cg
SVM	0.907	$C = 2.5$, kernel = poly, degree = 2
Decision Tree	0.896	criterion = entropy, leaf = 1

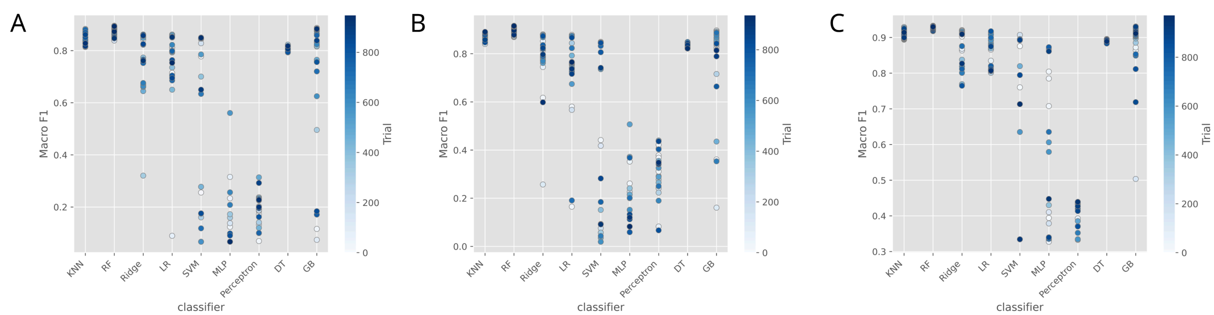

Figure S8: Phase 4 readout optimization slice plots across all three headline tasks. Each panel shows macro F1 vs. classifier family and its hyperparameters. (a) Balanced five-class: random forest achieves the highest macro F1 (0.895), slightly outperforming KNN (0.883) and gradient boosting (0.887). (b) Unbalanced five-class: random forest achieves the highest macro F1 (0.916 vs. 0.891 for KNN). (c) Binary: random forest (0.933) and gradient boosting (0.931) lead, with KNN (0.929) close behind.

8 Computational Scaling

Runtime scaled approximately quadratically with N prior to geometric cutoff. Table S16 reports per-stage wall-clock time per fold for each headline task (profiled on Host 1 with five parallel folds).

Table S16: Pipeline profiling for the three biological headline tasks plus the ESN baseline (five-fold cross-validation, using parallel processes on Host 1). "Train reservoir" and "Val reservoir" denote reservoir-state generation for training and test instances respectively. "Fit readout" is classifier training on aggregated states and "Predict" includes readout inference plus evaluation. All times are per fold (mean ± std across five folds).

Pipeline Stage	Balanced 5-class	Binary	Unbalanced 5-class	ESN baseline
Train Samples	3 212	30 172	12 885	3 212
Test Samples	803	7 542	3 222	803
Train reservoir	232.0 ± 1.7 s	2150.0 ± 2.0 s	931.4 ± 10.1 s	77.2 ± 1.1 s
Fit readout	0.05 ± 0.003 s	377.0 ± 2.0 s	0.19 ± 0.01 s	1.66 ± 0.08 s
Val reservoir	56.9 ± 0.4 s	536.0 ± 1.0 s	231.1 ± 1.7 s	19.27 ± 0.11 s
Predict	14.1 ± 0.3 s	33.0 ± 1.0 s	123.9 ± 4.2 s	0.165 ± 0.002 s
Per-instance sim.	≈ 72 ms (BioReservoir) ≈ 24 ms (ESN)			

DDE integration accounted for over 95% of biological pipeline wall time. The per-instance cost (≈72 ms for the BioReservoir, ≈24 ms for the ESN) scaled linearly with instance count.

9 Comparable Baselines

9.1 Leaky ESN Baseline

A leaky ESN (ReservoirPy) was optimized over 240 TPE trials on the balanced single-fold task (best: $F_1 = 0.879$, $N = 850$, `tanh`, $sr = 1.70$, $lr = 0.316$). A frozen-state readout study (1008 trials) selected ridge ($\alpha = 2.23 \times 10^{-4}$) as the best classifier and a final best-configuration five-fold rerun obtained $F_1 = 0.896 \pm 0.010$.

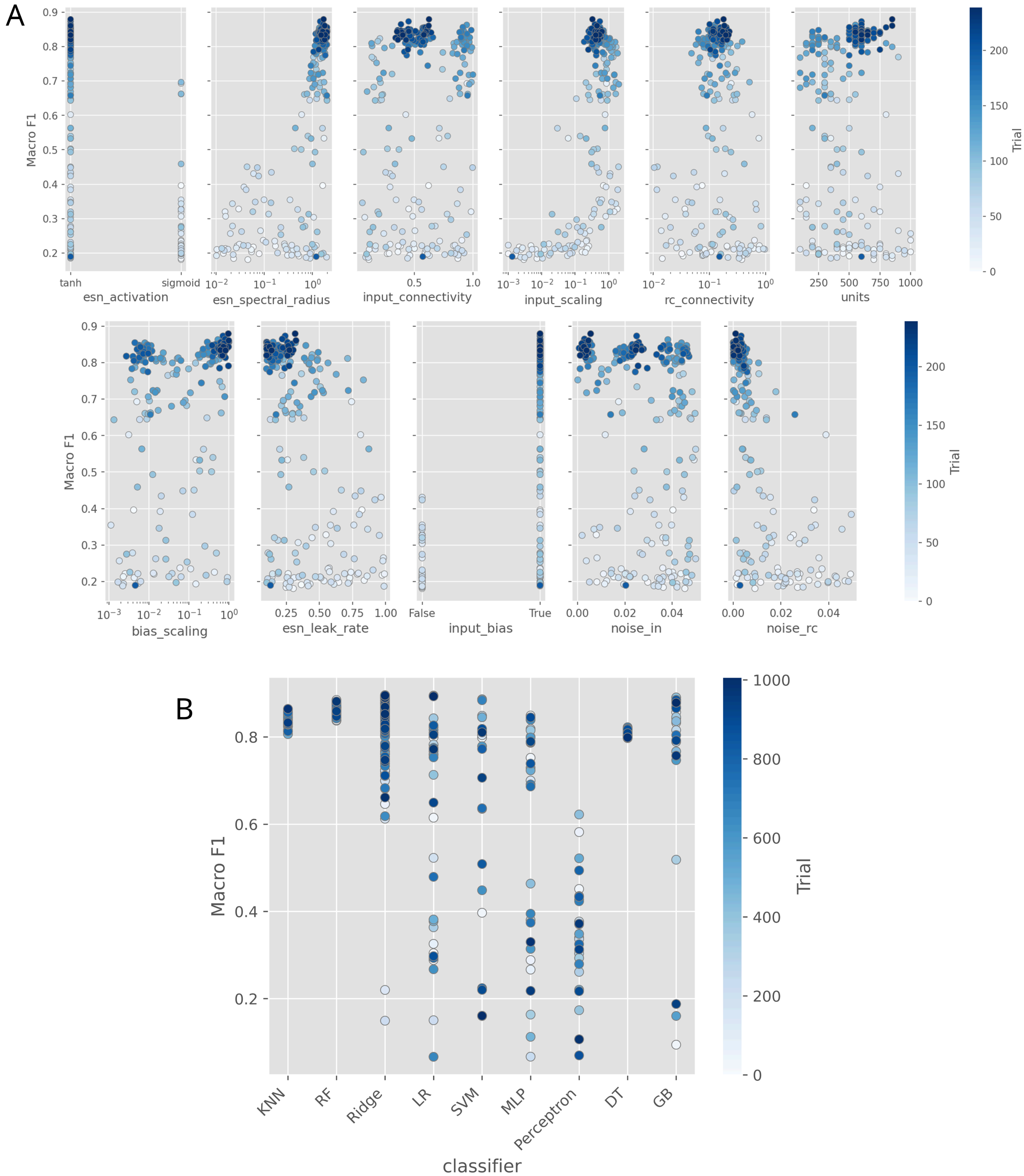

Figure S9: Slice plots for the two ESN comparison studies. (a) The ESN training study (`ESN-balanced`, 240 completed trials) shows that the top-performing region favors moderately large reservoirs, `tanh` activation, input bias, and intermediate recurrent connectivity. Clear patterns across many parameters suggest that echo-state networks are sensitive to specific hyperparameter choices. (b) The frozen-state readout study (`ESN-readout`, 1008 completed trials) is dominated by ridge configurations, but the only bad performers are MLP and Perceptron.

9.2 No-Reservoir Baseline

Nine classifier families (the seven Phase 4 families plus MLP and perceptron) were trained directly on raw z-scored 187 timestep ECG signals using the same five-fold CV protocol and macro-F1 objective. Three studies matched the Phase 4 task settings: B1-baseline (balanced, 4,015 instances), B2-baseline-unbalanced (5× configuration), and B3-baseline-binary (4,015-instance subset).

Table S17: Best macro F1-score per classifier across the three no-reservoir baseline studies (five-fold CV on raw z-scored ECG features). Bold entries mark the best classifier per task. The headline binary baseline ($F_1 = 0.964$) was obtained by rerunning the best KNN on the full 37,714-instance dataset.

Classifier	Balanced 5-class (480 trials)	Unbalanced 5-class (480 trials)	Binary (480 trials)
SVM	**0.886** (poly)	**0.915** (poly)	0.933 (RBF)
MLP	0.874	0.914	0.928
KNN	0.872	0.902	**0.935**
Gradient Boosting	0.869	0.895	0.919
Random Forest	0.868	0.908	0.935
Decision Tree	0.825	0.842	0.883
Logistic Regression	0.761	0.744	0.813
Ridge Classifier	0.736	0.737	0.813
Perceptron	0.693	0.668	0.738

9.3 Cross-Task Summary: Reservoir vs. Raw Features

The reservoir provides large F1 improvements for linear readouts (ridge, LR), negligible changes for instance-based methods (KNN, RF), and degradation for classifiers with built-in nonlinear mapping (SVM, MLP).

Table S18: Macro F1-score improvement (Δ) from reservoir states over raw features, per classifier and task. Positive values indicate the reservoir transformation improved performance; negative values indicate raw features were superior for that classifier. Classifiers are ordered by mean Δ across tasks.

	Balanced 5-class		Unbalanced 5-class		Binary	
Classifier	Δ	Raw	Δ	Raw	Δ	Raw
Ridge	+0.126	0.736	+0.143	0.737	+0.108	0.813
LR	+0.101	0.761	+0.134	0.744	+0.104	0.813
RF	+0.027	0.868	+0.008	0.908	−0.002	0.935
GB	+0.018	0.869	+0.002	0.895	+0.012	0.919
DT	−0.002	0.825	+0.007	0.842	+0.013	0.883
KNN	+0.011	0.872	−0.011	0.902	−0.006	0.935
SVM	−0.036	0.886	−0.066	0.915	−0.026	0.933
MLP	−0.313	0.874	−0.407	0.914	−0.055	0.928

10 Ablation Studies

Three ablation experiments and a topology comparison were conducted on the balanced five-class pipeline using the P2-distance best-trial parameters and the Phase 4 random forest readout. The main text (Section 2.8) presents the headline results and interpretation, per-fold values are reported in Table S19.

1. **Coupling ablation** ($\alpha_{\mathrm{rc}} = 0$): recurrent coupling removed so that oscillators evolve independently.
2. **Delay ablation** ($\tau = 0$): transcriptional delay removed, converting the DDE to an ODE system.
3. **Input scrambling**: input timesteps randomly permuted per instance, destroying temporal structure.

Table S19: Ablation results on balanced five-class classification (random forest readout, $n_{\mathrm{est}} = 326$, five-fold CV). The baseline is the full optimized model (P2-distance best trial).

Condition	Macro F1	Δ
Full model (baseline)	0.892 ± 0.013	—
No coupling ($\alpha_{\mathrm{rc}} = 0$)	0.844 ± 0.013	-0.048
No delay ($\tau = 0$)	0.793 ± 0.016	-0.099
Scrambled input	0.220 ± 0.020	-0.672

11 Cell-to-Cell Kinetic Heterogeneity

To assess robustness to biological noise, each reservoir node drew its 13 intracellular kinetic rate constants independently from log-normal distributions:

$$\theta_i \sim \theta_0 \cdot \exp\big(\mathcal{N}(0,\, \sigma^2)\big), \qquad \sigma = \sqrt{\ln(1 + \mathrm{CV}^2)}, \tag{3}$$

where θ_0 is the nominal value from Danino et al. (S1). Colony-level parameters were held fixed. All reservoir hyperparameters remained at P2-distance best-trial values with no re-tuning, testing inherent robustness. Eight CV levels (0–0.5) $\times$ five random seeds yielded 40 grid points, each evaluated with five-fold stratified cross-validation using the Phase 4 random forest readout.

11.1 Results

Performance was preserved or slightly improved at low-to-moderate heterogeneity (CV $\leq$ 0.15, mean $F_1 \geq 0.889$) relative to the homogeneous baseline ($F_1 = 0.889$). At higher CV, degradation was minimal: $F_1 = 0.889$ at CV = 0.2 and 0.885 at CV = 0.5 (-0.4%). Degradation concentrated on morphologically similar classes N and S (Table S20), while the most distinct class Q remained essentially invariant.

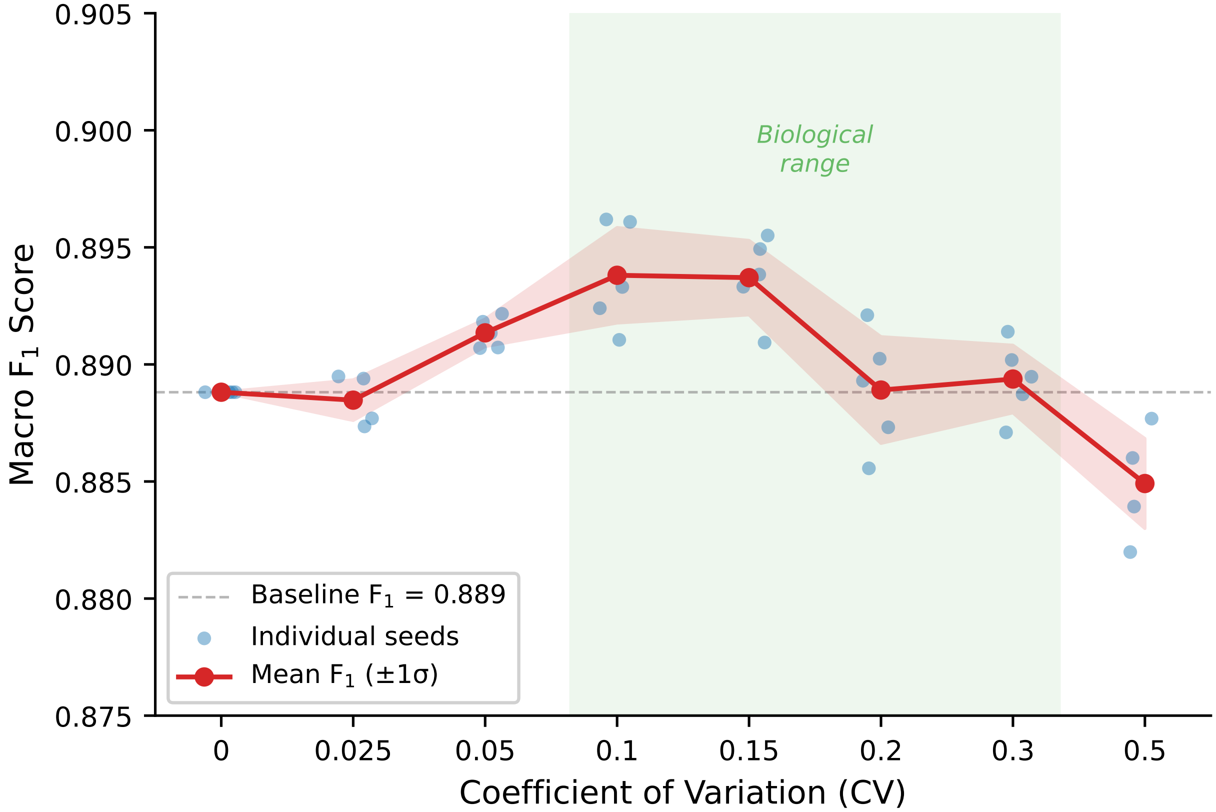

Figure S10: Macro F_1-score as a function of per-node kinetic parameter heterogeneity (coefficient of variation, CV). Red line: mean across five independent random seeds. Blue dots: individual seeds (each the mean of five CV folds). Green shading marks a representative low-to-moderate heterogeneity band (CV = 0.1–0.3). The dashed line indicates the homogeneous baseline ($F_1 = 0.889$). All trials used the hyperparameters optimized under homogeneous conditions with the Phase 4 random forest readout, with no re-tuning. Because the heterogeneity sweep uses a balanced evaluation set with equal class support, macro and weighted F_1 are numerically identical.

11.2 Per-Class Breakdown

Table S20: Per-class F_1-score (mean across seeds, five-fold CV with RF readout) as a function of kinetic-parameter heterogeneity. Class labels follow the AAMI standard: N (Normal), S (Supraventricular), V (Ventricular premature), F (Fusion), Q (Unknown/paced).

	Coefficient of Variation (CV)							
Class	0.0	0.025	0.05	0.1	0.15	0.2	0.3	0.5
N	0.822	0.817	0.820	0.824	0.823	0.820	0.823	0.819
S	0.854	0.855	0.860	0.863	0.864	0.858	0.860	0.854
V	0.897	0.899	0.903	0.904	0.903	0.897	0.898	0.893
F	0.912	0.910	0.915	0.915	0.913	0.907	0.905	0.901
Q	0.958	0.962	0.959	0.963	0.966	0.962	0.962	0.957
Macro	**0.889**	**0.888**	**0.891**	**0.894**	**0.894**	**0.889**	**0.889**	**0.885**

References

(S1) Danino, T., Mondragón-Palomino, O., Tsimring, L., and Hasty, J. (2010) A Synchronized Quorum of Genetic Clocks. *Nature 463*, 326–330.

(S2) Moody, G. B., and Mark, R. G. (2001) The Impact of the MIT-BIH Arrhythmia Database. *IEEE Engineering in Medicine and Biology Magazine 20*, 45–50.

(S3) Kachuee, M., Fazeli, S., and Sarrafzadeh, M. ECG Heartbeat Classification: A Deep Transferable Representation. 2018 IEEE International Conference on Healthcare Informatics (ICHI). 2018.

(S4) Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. Optuna: A Next-Generation Hyperparameter Optimization Framework. Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 2019; pp 2623–2631.